Patch preview: Results and appendix changes for the XYZ^2 paper

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This preview typesets the four Overleaf patches in `paper/overleaf_patches/` with the paper's preamble, macros and colours. The rest of the paper is not included, because the live Overleaf files could not be read from Perlmutter. References into those parts (Table IV, Sec. CSS map) point to placeholders here.

I. RELATION TO EARLIER XYZ^2 RESULTS

Patch `hw_relation.tex`: replaces the first two paragraphs of the subsection; the new table goes right after Table IV.

The circuits, noise models and decoders of our study and of that of Hetényi and Wootton differ (Table IV), so we compare rankings rather than values. They fit a plane to the threshold surface of a spin-qubit model with three error rates, a gate error, an idle error with bias $\eta_T = p_Z/(p_X + p_Y) = 20$ and a combined reset and readout error [1]. The intercepts of this plane are thresholds with only one kind of error switched on, so at the gate corner idle, reset and readout errors are all off. We measured the same corners in our own circuits with one error parameter p each, matching, $r = d$ rounds and finite-size fits over $d = 11$ to 21 (Table I). At the gate corner our thresholds are higher than theirs for every code, and in both studies the rotated surface code ranks above the XYZ^2 code. The ratio of the two thresholds is 0.770(17) in our circuits and 0.57 in theirs. The two studies rank the XZZX code differently. Our XZZX circuit uses native controlled Paulis and has the same detector error model as our CSS circuit at the gate corner, so the two codes reach the same threshold, 1.15(2)%. Their XZZX circuit adds single-qubit basis changes on the data qubits, each as noisy as a two-qubit gate, and reaches 0.37%. The honeycomb Floquet code reaches 0.88(1)% at the gate corner, the same as the 0.89(2)% of the XYZ^2 code, and the two codes also agree with the best of matching, correlated matching and belief-matching, 1.081(8)% against 1.10(4)%. Hetényi and Wootton place the XYZ^2 code slightly above the honeycomb code, with 0.465% against 0.43%.

We then added a depolarizing error of probability p on both qubits of every two-qubit gate, which models single-qubit gates as noisy as two-qubit gates. The XZZX threshold falls to 0.438(4)%, near the 0.37% of Hetényi and Wootton. Relative to our CSS threshold without these errors it falls to 0.380(6), against 0.45 for the ratio of their XZZX and CSS values. The noisy single-qubit gates therefore account for the low XZZX value of the spin-qubit study. When every code uses the same gates, the XZZX code ranks above the XYZ^2 code with and without single-qubit errors, as in the rest of our comparison. With single-qubit errors the XYZ^2 code also ranks above the honeycomb code, 0.326(8)% against 0.293(2)%. At the readout corner the honeycomb code has the lowest threshold in both studies, 1.007(7)% in ours against about 10% for the other three codes. With matching the XYZ^2 code reaches 10.6(3)% there, against 9.8(4)% for the CSS code, but with the best of our three decoders the XYZ^2 code and the surface codes agree within their errors. At the idle corner the XYZ^2 code ranks below the CSS code in our circuits, 0.88(1)% against 3.6(4)%, while it ranks above it in theirs, 4.2% against 3.94%. Our idle error acts on every qubit that waits in a gate layer, and the link ancillas of the XYZ^2 code wait in four of the six layers of a round. Their idle error acts only on the data qubits while the ancillas are read out and reset. With the idle error placed that way, the crossing of $d = 7$ and 9 is 10.5(6)% for the XYZ^2 code and 6.6(2)% for the CSS code, which restores their order.

	Gates			Idling		Readout	
Code	HW	Ours	Ours, 1q	HW	Ours	HW	Ours
Surface (CSS)	0.82	1.15(1)	0.433(3)	3.94	3.6(4)	14.5	9.8(4)
XZZX	0.37	1.15(2)	0.438(4)	15.1	9.8(3)	12.9	9.8(3)
XYZ^2	0.465	0.89(2)	0.326(8)	4.2	0.88(1)	16.5	10.6(3)
Honeycomb	0.43	0.88(1)	0.293(2)	1.16	0.392(3)	0.99	1.007(7)

TABLE I. Thresholds in percent at the corners of the threshold surface of Hetényi and Wootton, where only one kind of error is switched on. HW: the intercepts of their fitted plane [1], with $\eta_T = 20$, $d = 11$ to 17 and matching. Ours: matching, $r = d$ rounds, finite-size fits over $d = 11$ to 21 ($d = 12$ to 20 for the honeycomb code), weaker basis. Gates: a two-qubit depolarizing error of probability p after every two-qubit gate. Ours, 1q: in addition a depolarizing error of probability p on both qubits of every two-qubit gate. Idling: a Pauli error of total probability p with $p_Z/(p_X + p_Y) = 20$ on every qubit that waits in a gate layer and on the data qubits while the ancillas are reset and read out. Readout: every measurement outcome flipped with probability p.

II. RANKING

Patch `ranking.tex`: replaces the comment line "% Equal-qubit comparison"; adds Eq. (1) and the three-panel figure.

We also compare the codes below threshold at equal numbers of qubits, where the original proposals place the advantage of the XYZ^2 code. We run memories with $r = 3d$ rounds at $p = 10^{-3}$, 2×10^{-3} and 3×10^{-3} and convert the logical error rate p_L of each run into an error per round ϵ through

$$p_L = \tfrac{1}{2}\left[1 - (1 - 2\epsilon)^r\right]. \qquad (1)$$

For each code and noise model we use one decoder at every point, correlated matching, which gives the lowest error at the largest distances, and the weaker logical basis. The XYZ^2 code at $d = 5$, 9 and 11 has 99, 323 and 483 qubits, close to the 97, 337 and 449 qubits of the surface codes at $d = 7$, 13 and 15 (Fig. 1). Under SD6 at $p = 10^{-3}$ the error per round of the XYZ^2 code is 11(1) times that of the CSS code at 99 qubits and about 700 times, within a factor of two, at about 330 qubits. At $p = 2 \times 10^{-3}$ and 3×10^{-3} the factor grows from 9 and 7 at 99 qubits to 330 and 130 at about 470 qubits. The suppression factor $\Lambda = \epsilon(d)/\epsilon(d+2)$ at $p = 2 \times 10^{-3}$ is 4.0(2) for the XYZ^2 code against 6.6(1) for the CSS code, and each step in d costs the XYZ^2 code twice as many qubits. Under biased noise the XYZ^2 code beats the CSS code at equal qubits, by a factor of 3 to 5 at 99 qubits and 8 to 60 at the larger pairs for $\eta = 100$ and pure Z noise. The XZZX code beats the XYZ^2 code by a factor of 10 to 17 at 99 qubits and 80 to 850 at the larger pairs. With matching alone the XYZ^2 code also falls behind the CSS code under biased noise, so its gain over the CSS code relies on a decoder that reads the gauge level. At $p = 10^{-3}$ an error of 10^{-6} per round takes 244 qubits for the XYZ^2 code against 120 for the CSS code under SD6, and 134 against 192 for the CSS code and 82 for the XZZX code at $\eta = 100$. Taking ϵ from runs with $r = d$ and $r = 3d$, which removes the contribution of the time boundaries, changes it by -46% to $+17\%$ and changes none of these comparisons. Equal qubit counts therefore do not reverse the ranking of the thresholds. The XYZ^2 code gains on the CSS code under biased noise, and it stays behind the XZZX code below threshold as well as at threshold.

III. WHERE THE GAP COMES FROM

Patch `gap.tex`*: replaces the four paragraphs and the gap figure.*

We locate the gap to the XZZX code under SD6 with six variants of the noise model (Fig. 2). In the noiseless-link model, which we use only as a diagnostic, the link checks are noiseless. In the repeated-link models every link is measured twice or three times per round. In the hybrid model the links are EM3 pair measurements and the plaquettes keep SD6 gates. In the two idle-free models we remove every idle error, in the gate layers and while the ancillas are reset and read out, once with noisy and once with noiseless link checks.

The first diagnostic is noise on the link checks. With noiseless link checks the XYZ^2 memory threshold rises from 0.487(7)% to 0.626(8)%, against 0.813(20)% for the XZZX code, which closes 0.43(4) of the gap. Measuring every link twice per round gives 0.558(21)%, and measuring it three times gives 0.526(8)%, less than twice. Measuring the links as pair measurements raises the memory threshold to 0.575(6)%, still below the 0.81% of the surface codes. This hybrid circuit has fault distance d, which we checked at $d = 3$, because a faulty link measurement puts a two-qubit error on one block.

The second diagnostic is idling. Each bulk data qubit takes part in one link and three hexagons, so it is busy in four of the six gate layers and idles in the other two. Without idle errors the XYZ^2 memory threshold rises from 0.487(7)% to 0.854(6)%, above the 0.813(20)% of the XZZX code under SD6. The XZZX code also gains from removing its idle errors, to 1.103(9)%. Measured against this reference, the idle errors account for 0.60(1) of the gap and the noise on the link checks for 0.23(2). With noiseless link checks and without idle errors the XYZ^2 threshold is 1.12(3)%, which closes 1.02(5) of the gap. Matching alone gives the same picture, with 0.691(6)% and 0.99(1)% for the two idle-free XYZ^2 models against 0.93(2)% for the XZZX code without idle errors. No schedule or compilation that we tried closes the gap under SD6.

The map of Sec. [CSS map] shows the structural cost. Under depolarizing noise the XYZ^2 code is a rotated surface code concatenated with two-qubit repetition codes. It uses $4d^2 - 1$ qubits with gates, against $2d^2 - 1$ for the surface codes at the same distance, and every hexagon has weight six. Criger and Terhal found that the circuit-level threshold of the $[[4, 2, 2]]$-concatenated toric code depends on the schedule of its weight-eight checks, and their shortest schedule reached about 0.41%, against about 0.6% for the toric code [2]. In our circuits the combined diagnostic leaves none of the threshold gap, within its error, for the weight-six hexagons and the doubled qubit count. Their cost appears below threshold, where the XYZ^2 code needs more qubits than the surface codes for the same error per round under SD6 (Sec. II).

IV. APPENDIX: FITS AND SAMPLING

Patch `app_fits.tex`*: two sentences added at the end of the paragraph on sampling.*

The sub-threshold runs of Sec. II stop each point at 100 logical errors, 300 at the equal-qubit pairs, or at 10^9 shots, within at most 4 core-hours for matching and correlated matching and 8 for belief-matching, twice that at the pairs, split into replicas of at most two hours. We plan these budgets from short calibration runs and measured decoding times, end a series after the first distance with fewer than 10 errors at its cap, and treat a point without errors in N shots as the 95% upper bound $3/N$ on its logical error rate per shot.

[1] B. Hetényi and J. R. Wootton, Tailoring quantum error correction to spin qubits, Phys. Rev. A **109**, 032433 (2024), 2306.17786.

[2] B. Criger and B. Terhal, Noise thresholds for the [[4,2,2]]-concatenated toric code, Quantum Inf. Comput. **16**, 1261 (2016), 1604.04062.

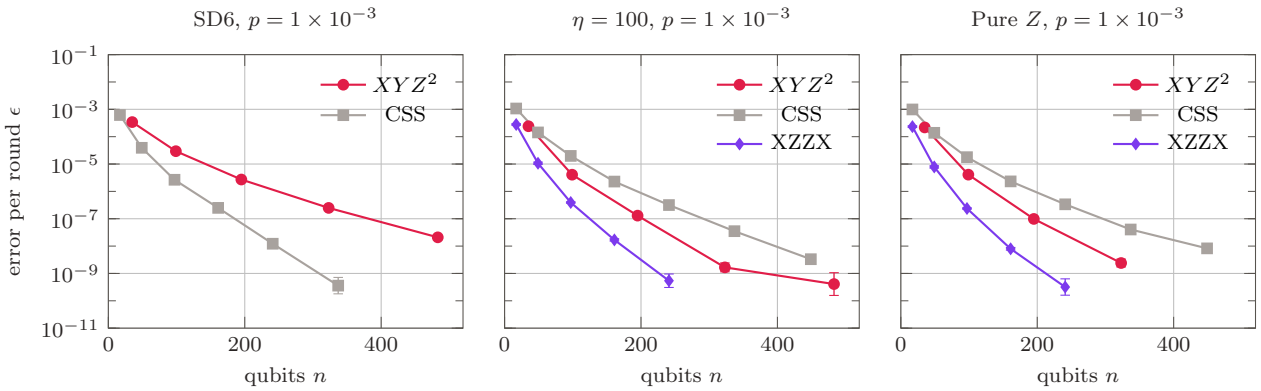


FIG. 1. Error per round ϵ of Eq. (1) against the number of qubits n at $p = 10^{-3}$ and $r = 3d$, with correlated matching and the weaker logical basis of each code, under SD6 (left), Z-biased noise with $\eta = 100$ (middle) and pure Z noise (right). Error bars are 68% Wilson intervals. Open triangles are 95% upper bounds where a run saw no logical error. Under SD6 the CSS circuit stands for the XZZX and XY circuits, which share its detector error model.

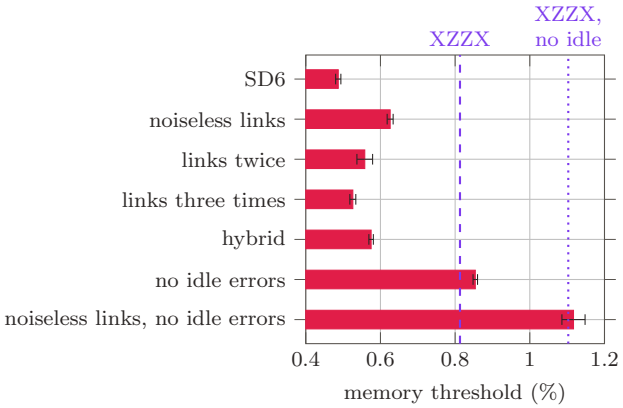


FIG. 2. Memory thresholds of the XYZ^2 code under SD6 and six variants of it: the best of matching, correlated matching and belief-matching in the weaker logical basis, $r = d$ rounds, finite-size fits over $d = 11$ to 21. The noiseless-link variants are diagnostics, not hardware. In the idle-free variants no qubit has an idle error, in the gate layers or while the ancillas are reset and read out. The dashed line marks the XZZX code under SD6 and the dotted line the XZZX code without idle errors.